ties, industry, government, and nongovernmental research or health organizations. Those with master's degrees generally start as project officers or staff members, coordinating data collection and analysis for health studies and as they grow in experience they will advance within their organizations. The doctoral degree often leads to academic careers on university faculty and leadership positions in other research organizations and industry. Physicians who earn master's or doctoral degrees in public health also often fill these positions. Epidemiologists who receive additional training (usually as doctoral or postdoctoral students) in human and statistical genetics are often called "genetic epidemiologists," in recognition of their specialty within epidemiology. Compensation varies widely, depending on the level of education, employment setting, and experience. In 2001 the starting salary for a new graduate with a master's degree and no previous work experience might be in the $30,000 to $40,000 range; a Ph.D. or M.D./M.P.H. with several years' experience might earn over $100,000 in industry.

Many Roles, Many Rewards

Epidemiologists interested in the influence of genetic factors on health may play several types of roles on a research project, including assisting in the overall design of the study, developing instruments to collect nongenetic risk-factor data, and using that data to investigate possible interactions between genetic and nongenetic (environmental) factors that influence health. Genetic epidemiologists perform a similar role, using study designs and statistical approaches developed specifically for the analysis of human genetics data.

The professional rewards of a career in epidemiology are the excitement of discovery and the knowledge that epidemiologic studies can be used to help people improve or maintain their health over time. Epidemiologic research has significantly improved the public's health over the past century. Research results have been used to identify new medicines to treat disease, to educate the public about the health effects of cigarette smoking and inactive lifestyles, and to improve sanitation and water treatment, significantly reducing the burden of infectious disease in heavily populated areas. SEE ALSO Gene and Environment; Population Screening; Public Health, Genetic Techniques in; Statistical Geneticist.

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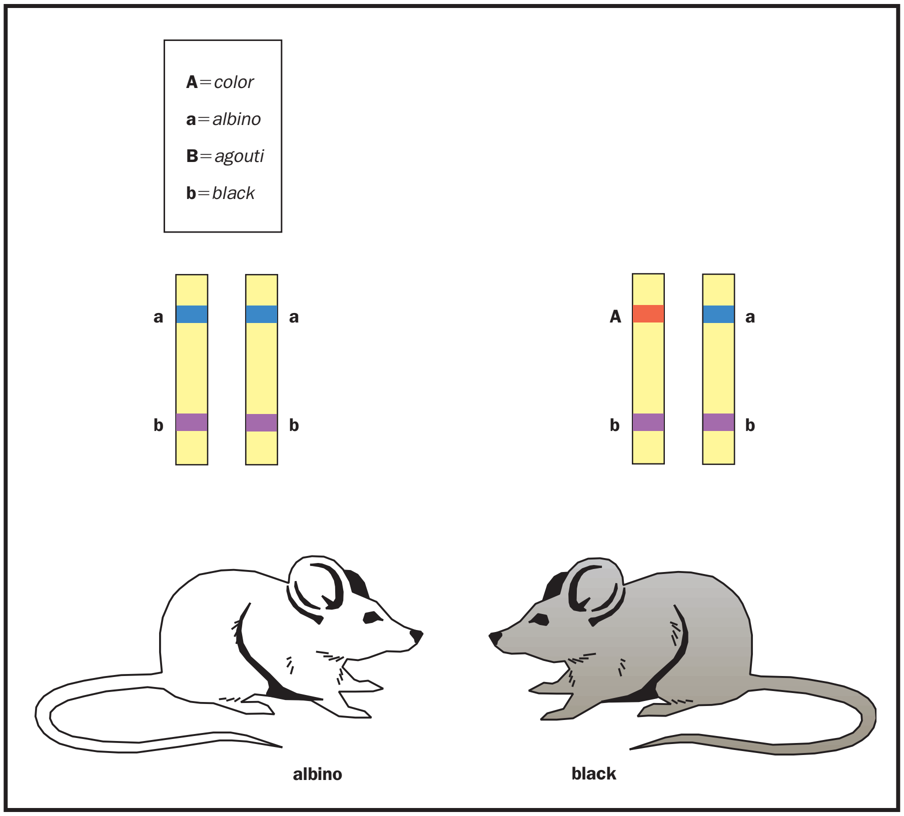
Epistasis

Epistasis, first defined by the English geneticist William Bateson in 1907, is the masking of the expression of a gene at one position in a chromosome, or **locus**, at one or more genes at other positions. Epistasis should not be confused with dominance, which refers to the interaction of genes at the same locus. The human genome contains from 30,000 to 70,000 gene loci. Some of them are involved in numerous interactions, making it difficult to identify their role in development and metabolism. As we learn more about

locus site on a chromosome (plural, loci)

The "A" locus is epistatic to the "B" locus. The "B" locus can influence coat color only if there is at least one dominant "A" allele at the "A" locus.



the human and other genomes, it becomes clear that the borrowed phrase "no gene is an island" is an appropriate expression to describe the interplay among gene loci.

Puzzling Inheritance Patterns Explained

phenotype observable characteristics of an organism

antigen a foreign substance that provokes an immune response

dominant controlling the phenotype when one allele is present

homozygous containing two identical copies of a particular gene

There are many examples of epistasis. One of the first to be described in humans is the Bombay **phenotype**, involving the ABO blood group system. Individuals with this phenotype lack a protein called the H **antigen** (genotype hh), which is used to form A and B antigens. Even though such individuals may have A or B genes, they appear to be blood group O because they lack the H antigen.

Another well-known example is coat color in mice. Two coat-color loci are involved. At locus A, color is **dominant** over albino (lack of pigment). At locus B, the coat color agouti is dominant over black. A mouse that is **homozygous** for the albino gene will show no pigment regardless of its genotype at the other locus. Thus the A and B loci are epistatic.

It is likely that the phenomenon of lack of penetrance, in which a dominant gene fails to be expressed, is often due to epistasis. There are many cases where dominant disorders, such as polydactyly (in which individuals have extra fingers or toes), appear to "skip generations." The nonexpression of the dominant gene is likely due to the alleles the individual has at an independent locus that is epistatic to the polydactyly locus. Lack of penetrance may also be accompanied by variable expressivity, where a gene is only partially expressed. As the molecular basis of these disorders becomes known, the reason for nonpenetrance will be easier to determine.

Such interactions between loci probably occur in the genetic etiology of complex traits such as the psychiatric disorders schizophrenia and manic depression. David Lykken, a genetic psychologist at the University of Minnesota, coined the term "emergenesis" to describe multiple gene interactions involved in a specific complex trait. After comparing EEG (electroencephalogram, or "brain wave") data from identical and fraternal twins, Lykken concluded that multiple-level interactions of independent or partly independent genes must be involved.

Epistatic interactions make it difficult to identify loci conferring risk for complex disorders, and they may be a major reason that researchers have made only slow progress in mapping susceptibility genes for complex disorders. To locate interacting loci involved in the genetic origins of complex diseases requires collecting DNA samples from a large number of families where two or more individuals have the disorder. Such large-scale studies are usually difficult to conduct.

Interactions among Proteins

As the Bombay phenotype demonstrates, it is actually proteins, not the genes, that interact. After identifying interacting loci, the next step is determining the proteins that the genes at those loci encode, and the properties of those proteins.

The emerging field that involves the study of proteins and protein interactions is called proteomics. New techniques are now available to locate proteins that interact with one another. In the yeast two-hybrid system, one such technique, one protein is used as bait, and a pool of unknown proteins, referred to as prey proteins, are tested to see if any of them bind to the bait. Binding, if it occurs, triggers a reaction that causes yeast cells to turn blue. In one experiment testing a protein's interactions with more than 1,000 other proteins, 950 interactions were found. Not all of these interactions are likely to occur or be important in the organism, but such results indicate how common, and complex, protein interactions are in living organisms. SEE ALSO Blood Type; Complex Traits; Inheritance Patterns; Proteomics; Psychiatric Disorders.

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Escherichia coli

Escherichia coli (*E. coli*) is a very common bacterium that normally inhabits the digestive tract of animals, including humans. It is widespread in the natural world and can also be found in soil and water. It is a member of the

Escherichia coli

bacterial family Enterobacteriaciae, which also includes the bacteria *Shigella*, *Salmonella*, and *Yersinia*, among others. Some of these organisms, including *E. coli*, can cause serious diseases under certain conditions.

Attributes of *E. coli*

E. coli is important to human health because it is a source of vitamins B12 and K, which it manufactures from undigested food in the large intestine. Unlike many other intestinal bacteria, *E. coli* can survive and grow in the presence of oxygen (although it can also grow without oxygen), which makes it a useful experimental model organism in the laboratory.

Even though *E. coli* is a single species of bacteria, many different varieties (called strains) of the species exist. Each has different characteristics, and while some are safe model organisms, others can cause potentially deadly disease. This is the case with *E. coli* 0157:H7, which is considered a dangerous **pathogen** which can infect humans. This strain is significantly different from the commonly used laboratory strains, which do not cause disease.

pathogen disease-causing organism

Importance in Laboratory Studies

E. coli is the most well-understood bacterium in the world, and is an extremely important model organism in many fields of research, particularly molecular biology, genetics, and biochemistry. It is easy to grow under laboratory conditions, and research strains are very safe to work with. As with many bacteria, *E. coli* grows quickly, which allows many generations to be studied in a short time. In fact, under ideal conditions, *E. coli* cells can double in number after only 20 minutes.

Furthermore, a very large number of *E. coli* bacteria can be grown in a small space—many millions in a drop of broth, for example. These are important characteristics in genetic experiments, which often involve selecting a single bacterial cell from among millions of candidates, then allowing it to reproduce into high numbers again to perform additional experiments.

Many vital techniques, such as molecular cloning and overexpression of cloned genes, were initially developed in *E. coli* and are still simpler and more effective in the bacterium. Crucial experiments that illuminated the details of fundamental biological processes such as DNA replication, **transcription**, and **translation** were performed for the first time or with greatest success in *E. coli*. The bacterium is still a primary resource in many modern laboratories. Even research efforts that focus on other organisms, including humans or crop plants, often use *E. coli* extensively as a tool to facilitate cloning and DNA sequencing.

transcription messenger RNA formation from a DNA sequence

translation synthesis of protein using mRNA code

Discoveries Made in *E. coli*

Some of the discoveries made in *E. coli* have provided an invaluable framework for understanding biological processes in more complex organisms. As mentioned above, many fundamental processes that are shared by all living things are most easily studied in this simple bacterial model. Furthermore, *E. coli* has served as a model for understanding the biology of other bacteria.

The ways in which *E. coli* interacts with the human body are in many cases very similar to the ways that other disease-causing organisms act. Therefore, this model organism has been important in the study of human health, and has allowed researchers to ask questions about bacteria in general (for example, how antibiotics stop infections, or how the immune system fights off disease).

Genome Sequenced Early

Sequencing of the *E. coli* K-12 strain **genome** (a popular model strain) was completed in 1997; subsequently, at least two collections of the pathogenic 0157:H7 strain have been completely sequenced. The bacterium has a genome of approximately 4.3 million base pairs of DNA, and carries about 4,400 genes. Interestingly, only about 50 percent of the predicted genes have been described and characterized, a surprisingly low percentage for such a well-understood organism. For this and other reasons, *E. coli* remains one of the most significant model organisms used today. SEE ALSO Chromosome, Prokaryotic; Eubacteria; Genome; Human Genome Project; Model Organisms; Plasmid.

genome the total genetic material in a cell or organism

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Eubacteria

The Eubacteria, also called just "bacteria," are one of the three main domains of life, along with the Archaea and the Eukarya. Eubacteria are prokaryotic, meaning their cells do not have defined, membrane-limited nuclei. As a group they display an impressive range of biochemical diversity, and their numerous members are found in every habitat on Earth. Eubacteria are responsible for many human diseases, but also help maintain health and form vital parts of all of Earth's ecosystems.

Structure

Like archeans, eubacteria are prokaryotes, meaning their cells do not have nuclei in which their DNA is stored. This distinguishes both groups from the eukaryotes, whose DNA is contained in a nucleus. Despite this structural resemblance, the Eubacteria are not closely related to the Archaea, as shown by analysis of their RNA (see below).

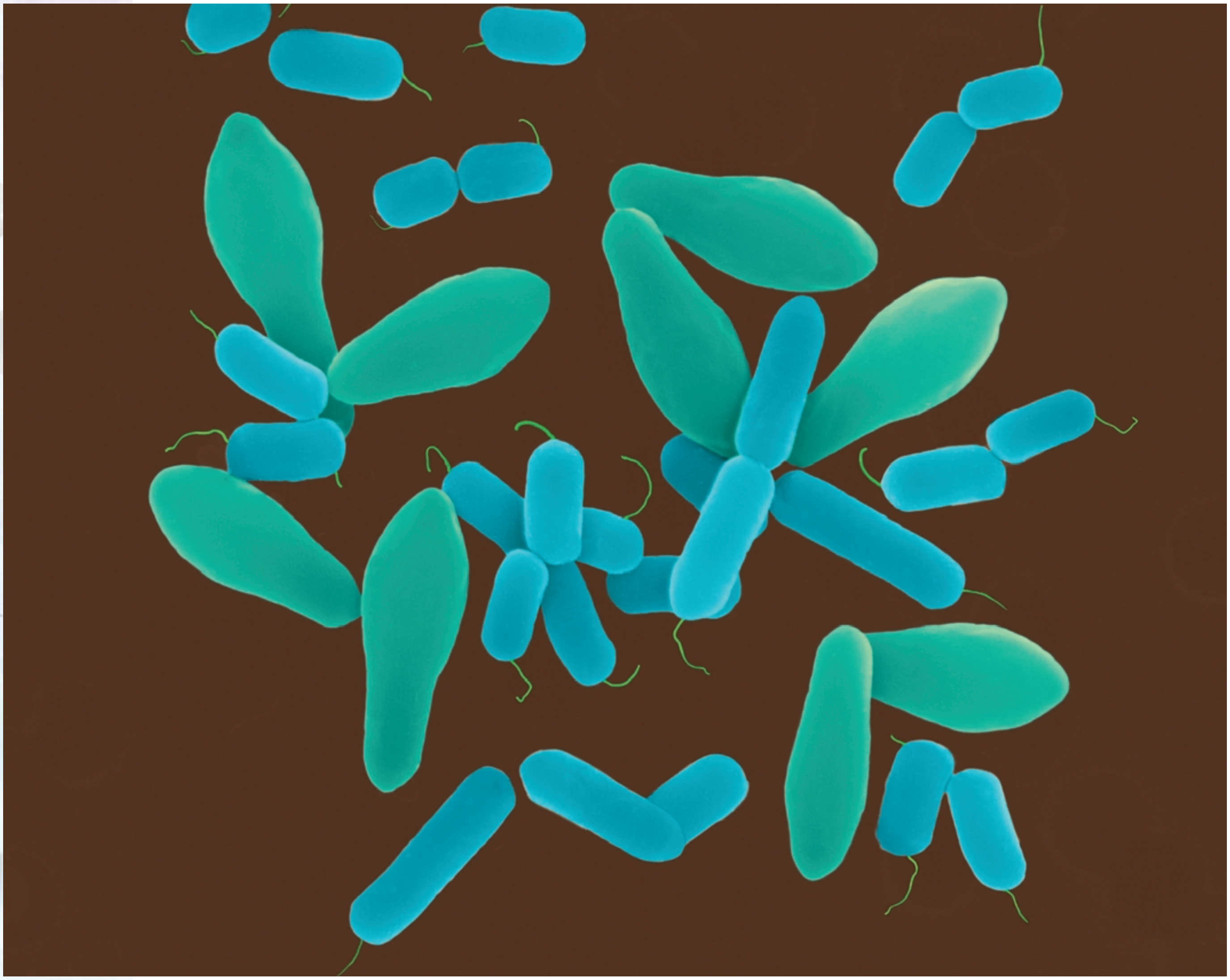
Eubacteria are enclosed by a cell wall. The wall is made of cross-linked chains of peptidoglycan, a **polymer** that combines both amino acid and sugar chains. The network structure gives the wall the strength it needs to maintain its size and shape in the face of changing chemical and **osmotic** differences outside the cell. Penicillin and related antibiotics prevent bacterial cell growth by inactivating an enzyme that builds the cell wall. Penicillin-resistant bacteria contain an enzyme that chemically modifies penicillin, making it ineffective.

polymer molecule composed of many similar parts

osmotic related to differences in concentrations of dissolved substances across a permeable membrane

Some types of bacteria have an additional layer outside the cell wall. This layer is made from lipopolysaccharide (LPS), a combination of **lipids** and sugars. There are several consequences to possessing this outer layer. Of least import to the bacteria but significant for researchers, this layer prevents

lipids fat or waxlike molecules, insoluble in water



Magnified 1,600 times, a scanning electron micrograph of *Clostridium perfringens* captures a Gram-positive, rod-shaped bacterium forming endospores. This bacterium causes food poisoning, wound infections, and gas gangrene.

them from retaining a particular dye (called Gram stain) that is used to classify bacteria. Bacteria that have this LPS layer are called Gram-negative, in contrast to Gram-positive bacteria, which do not have an outer LPS layer and which do retain the stain. Of more importance to both the bacteria and the organisms they infect is that one portion of the LPS layer, called endotoxin, is particularly toxic to humans and other mammals. Endotoxin is partly to blame for the damage done by infection from *Salmonella* and other Gram-negative species.

Within the cell wall is the plasma membrane, which, like the eukaryotic plasma membrane, is a phospholipid bilayer studded with proteins. Embedded in the membrane and extending to the outside may be flagella, which are whiplike protein filaments. Powered by molecular motors at their base, these spin rapidly, propelling the bacterium through its environment.

Within the plasma membrane is the bacterial cytoplasm. Unlike eukaryotes, bacteria do not have any membrane-bound organelles, such as mitochondria or chloroplasts. In fact, these two organelles are believed to have evolved from eubacteria that took up residence inside an ancestral eukaryote.

Bacterial cells take on one of several common shapes, which until recently were used as a basis of classification. Bacilli are rod shaped; cocci are spherical; and spirilli are spiral or wavyshaped. After division, bacterial cells may remain linked, and these form a variety of other shapes, from clusters to filaments to tight coils.

Metabolism

Despite the lack of internal compartmentalization, bacterial metabolism is complex, and is far more diverse than eukaryotic metabolism. Within the Eubacteria there are species that perform virtually every biochemical reaction known (and much bacterial chemistry remains to be discovered). Most of the vitamins humans require in our diet can be synthesized by bacteria, including the vitamin K humans absorb from the *Escherichia coli* (**E. coli**) bacteria in our large intestines.

E. coli common bacterium of the human gut, used in research as a model organism

The broadest and most significant metabolic distinction among the Eubacteria is based on the source of energy they use to power their metabolism. Like humans, many bacteria are heterotrophs, consuming organic (carbon-containing) high-energy compounds made by other organisms. Other bacteria are chemolithotrophs, which use inorganic high-energy compounds, such as hydrogen gas, ammonia, or hydrogen sulfide. Still others are phototrophs, using sunlight to turn simple low-energy compounds into high-energy ones, which they then consume internally.

For all organisms, extraction of energy from high-energy compounds requires a chemical reaction in which electrons move from atoms that bind them loosely to atoms that bind them tightly. The difference in binding energy is the profit available for powering other cell processes. In almost all eukaryotes, the ultimate electron acceptor is oxygen, and water and carbon dioxide are the final waste products. Some bacteria use oxygen for this purpose as well. Others use sulfur (forming hydrogen sulfide, which has a strong odor), carbon (forming flammable methane, common in swamps), and a variety of other compounds.

Bacteria that use oxygen are called aerobes. Those that do not are called anaerobes. This distinction is not absolute, however, since many organisms can switch between the two modes of metabolism, and others can tolerate the presence of oxygen even if they do not use it. Some bacteria die in oxygen, however, including members of the **Gram positive** *Clostridium* genus. *Clostridium botulinum* produces botulinum toxin, the deadliest substance known. *C. tetani* produces tetanus toxin, responsible for tetanus and "lockjaw," while other *Clostridium* species cause gangrene.

Gram positive able to take up Gram stain, used to classify bacteria

Life Cycle

When provided with adequate nutrients at a suitable temperature and pH, *E. coli* bacteria can double in number within 20 minutes. This is faster than most species grow, and faster than *E. coli* grows under natural conditions. Regardless of the rate, the growth of a bacterium involves synthesizing double the quantity of all its parts, including membrane, proteins, **ribosomes**, and DNA. Separation of daughter cells, called binary fission, is accomplished by creating a wall between the two halves. The new cells may eventually separate, or may remain joined.

ribosomes protein-RNA complexes at which protein synthesis occurs

Scanning electron micrograph of bacteria in rod and cocci form.

When environmental conditions are harsh, some species (including members of the genus *Clostridium*) can form a special resistive structure within themselves called an endospore. The endospore contains DNA, ribosomes, and other structures needed for life, but is metabolically inactive. It has a protective outer coat and very low water content, which help it survive heating, freezing, radiation, and chemical attack. Endospores are known to have survived for several thousand years, and may be capable of surviving for much longer, possibly millions of years. When exposed to the right conditions (presence of warmth and nutrients), the endospore quickly undergoes conversion back into an active bacterial cell.

DNA

plasmids small rings of DNA found in many bacteria

nucleoid region of the bacterial cell in which DNA is located

Most eubacteria have DNA that is present in a single large circular chromosome. In addition, there may be numerous much smaller circles, called **plasmids**. Plasmids usually carry one or a few genes. These often are for specialized functions, such as metabolism of a particular nutrient or antibiotic.

Despite the absence of a nucleus, the chromosome is usually confined to a small region of the cell, called the **nucleoid**, and is attached to the inner membrane. The bacterial genome is smaller than that of a eukaryote. For example, *E. coli* has only 4.6 million base pairs of DNA, versus three billion in humans. As in eukaryotes, the DNA is tightly coiled to fit it into the cell. Unlike eukaryotes, however, the DNA is not attached to histone proteins.

Much of what we know about DNA replication has come from study of bacteria, particularly *E. coli*, and the details of this process are discussed elsewhere in this encyclopedia. Unlike eukaryotic replication, prokaryotic replication begins at a single point, and proceeds around the circle in both directions. The result is two circular chromosomes, which are separated during cell division. Plasmids replicate by a similar process.

Gene Transfer

While bacteria do not have sex like multicellular organisms, there are several processes by which they obtain new genes: **conjugation**, transformation, and transduction. Conjugation can occur between two appropriate bacterial strains when one (or both) extends hairlike projections called pili to contact the other. The chromosome, or part of one, may be transferred from one bacterium to the other. In addition, plasmids can be exchanged through these pili. Some bacteria can take up DNA from the environment, a process called transformation. The DNA can then be incorporated into the host chromosome.

conjugation a type of DNA exchange between bacteria

Some bacterial viruses, called phages, can carry out transduction. With some phages, the virus temporarily integrates into the host chromosome. When it releases itself, it may carry some part of the host DNA with it. When it goes on to infect another cell, this extra DNA may be left behind in the next round of integration and release. Other phages, called generalized transducers, package fragments of the chromosome into the phage instead of their own **genome**. When the transducing phage infects a new cell, they inject bacterial DNA. These phages lack their own genome and are unable to replicate in the new cell. The inserted bacterial DNA may recombine (join in with) the host bacterial chromosome.

genome the total genetic material in a cell or organism

Gene Regulation and Protein Synthesis

Gene expression in many bacteria is regulated through the existence of operons. An operon is a cluster of genes whose protein products have related functions. For instance, the *lac* operon includes one gene that transports lactose sugar into the cell and another that breaks it into two parts. These genes are under the control of the same **promoter**, and so are transcribed and translated into protein at the same time. RNA polymerase can only reach the promoter if a repressor is not blocking it; the *lac* repressor is dislodged by lactose. In this way, the bacterium uses its resources to make lactose-digesting **enzymes** only when lactose is available.

promoter DNA sequence to which RNA polymerase binds to begin transcription

enzymes proteins that control a reaction in a cell

Other genes are expressed constantly at low levels; their protein products are required for "housekeeping" functions such as membrane synthesis and DNA repair. One such enzyme is DNA gyrase, which relieves strain in the double helix during replication and repair. DNA gyrase is the target for the antibiotic ciproflaxin (sold under the name Cipro), effective against *Bacillus anthracis*, the cause of anthrax. Since eukaryotes do not have this type of DNA gyrase, they are not harmed by the action of this antibiotic.

As in eukaryotes, translation (protein synthesis) occurs on the ribosome. Without a nucleus to exclude it, the ribosome can attach to the messenger RNA even while the RNA is still attached to the DNA. Multiple ribosomes can attach to the same mRNA, making multiple copies of the same protein.

The ribosomes of eubacteria are similar in structure to those in eukaryotes and archaea, but differ in molecular detail. This has two important consequences. First, sequencing ribosomal RNA molecules is a useful tool for understanding the evolutionary diversification of the Eubacteria. Organisms with more similar sequences are presumed to be more closely related. The same tool has been used to show that Archaea and Eubacteria are not closely related, despite their outward similarities. Indeed, Archaea are more closely related to eukaryotes (including humans) than they are to Eubacteria.

Second, the differences between bacterial and eukaryotic ribosomes can be exploited in designing antibacterial therapies. Various unique parts of the bacterial ribosome are the targets for numerous antibiotics, including streptomycin, tetracycline, and erythromycin. SEE ALSO ARCHAEA; CELL, EUKARYOTIC; CHROMOSOME, PROKARYOTIC; CONJUGATION; *ESCHERICHIA COLI*; OPERON; PLASMID; TRANSDUCTION; TRANSFORMATION.

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Eugenics

While the idea of improving humans through selective breeding is at least as old as the ancient Greeks, it gained widespread prominence after 1869. In 1883, Sir Francis Galton coined the word "eugenics," from the Greek word *eugenes*, meaning "well-born" or "hereditarily endowed with noble qualities," to describe this new science of directed human evolution. Galton's work, and the subsequent rediscovery of Gregor Mendel's genetic studies, convinced many scientists and social reformers that eugenic control over heredity could improve human life.

Galton's ideas swept America during the Progressive Era of the early twentieth century. At that time, many scientists and laypeople believed that eugenics could facilitate social progress by eradicating problems ranging from alcoholism and prostitution to poverty and disease. What better way to prevent such misfortunes, eugenicists asked, than to prevent the birth of people genetically susceptible to them? Eugenics seemed to offer an efficient and humane solution to society's ills. Unfounded hope in this imperfect science, however, ultimately contributed to repressive social policies, including marriage and immigration restriction, forced sterilization, segregation, and, in the case of Nazi Germany, euthanasia ("mercy killing") and genocide, all in the name of human betterment.

British Origins

Charles Darwin's theories of evolution by natural selection rocked the scientific world in 1859, and prompted his cousin, Galton, to study human evolution. Galton's first book, *Hereditary Genius* (1869), analyzed famous European families and concluded that "genius," which he defined as the ability to succeed in life, tended to run in families. Galton believed that individuals inherited the traits that destined them to either success or failure. Thus, success resulted from biology, not from the wealth or poverty of a person's background, and controlled breeding might permanently improve the human race.

Galton hoped to speed and direct human evolution. Writing in *Inquiries into the Human Faculty and Its Development* (1883), Galton defined eugenics as "the science of improving stock . . . to give the more suitable races or strains of blood a better chance of prevailing speedily over the less suitable than they otherwise would have had." Familiar with farmers' achievements in breeding more-valuable plants and animals, Galton believed that such methods were "equally applicable to men, brutes, and plants."

Galton identified those fit folk who should have children and stigmatized those he deemed unfit for parenthood. He also believed then-accepted notions of "racial" superiority and inferiority, had more to do with class and cultural prejudice than with biological difference. Galton assumed that wealthy people like himself were fit, whereas poor folk were unfit. Northern European "white" people stood atop the evolutionary scale of fitness, followed by "whites" from southeast Europe, Asians, Native Americans, Africans, and Australian Aborigines.

Positive and Negative Eugenics

Galton identified positive and negative eugenics as the two basic methods to improve humanity. Positive eugenics used education, tax incentives, and childbirth stipends to encourage **procreation** among fit people. Education would convince fit parents to have more children, out of a desire to increase the common good. Lower taxes on larger families and the provision of a small birth payment for each "eugenic" child would provide further inducements. Conversely, eugenically educated but unfit people would selflessly forgo procreation, to prevent the propagation of their hereditary "taint." Believing that neither altruism nor self-interest would be enough to control the unfit, however, many eugenicists also advocated negative eugenics.

procreation reproduction

Negative eugenics sought to limit procreation through marriage restriction, segregation, sexual sterilization, and, in its most extreme form, euthanasia. In an attempt to decrease procreation among the "unfit," laws prohibited marriage to people with diseases, or other conditions believed to be hereditary. Similar restrictions banned marriage between people of different races, in order to prevent **miscegenation**. Popular in the United States, antimiscegenation laws sought to use science to legitimize racial prejudice. Since marriage restriction failed to stop extramarital procreation, eugenicists argued for more intrusive interventions.

miscegenation racial mixing

Many of these more intrusive interventions relied upon segregation. For example, individuals judged unfit might be segregated in institutions such as insane asylums, tuberculosis sanatoriums, and homes for the so-called feebleminded or mentally retarded. Isolated from "normal" society, these people were also segregated by sex within the institution to prevent procreation. Segregation through incarceration, however, proved too costly to be applied to all but the most severely handicapped.

Compulsory sexual sterilization of those individuals deemed "feebleminded" or "slow" promised eugenic and economic benefits for society. Once sterilized, such individuals posed no eugenic risk; sexual intercourse would never result in pregnancy. Sterilized individuals could therefore return to society and work, rather than remaining an economic "burden" in an institution. Many social reformers argued that compulsory sterilization was more humane than locking people away during their childbearing years.

In the case of individuals afflicted with gross physical or mental abnormalities, the most radical eugenic intervention was proposed: euthanasia. While many eugenicists theorized about euthanasia, very few seriously considered it as a real possibility. This would change with the advent of Nazi eugenics in Germany.

Mendelian Inheritance, Intelligence Testing, and American Eugenics

Galton's eugenic ideas found fertile ground in America after 1900, when scientists rediscovered Mendel's findings regarding the inheritance of physical traits in pea plants. Mendel's notions of "dominant" and "recessive" genetic traits, easily identified in "lower" organisms such as plants and animals, convinced people that human eugenic improvement was possible. Scientists assumed that even complex human traits such as intelligence and behavior behaved as simple genetic "unit characters," such as height or color in peas.

The advent of intelligence testing in the 1900s provided a new way to quantify Galton's notion of genius. American eugenicists assessed an individual's eugenic worth by combining his intelligence quotient (IQ) with a Galtonian study of the family pedigree. Psychologist Henry Herbert Goddard published one famous study, *The Kallikak Family*, in 1912. Goddard traced two family lines that originated with a common male ancestor, whom he called Martin Kallikak (from the Greek words for beautiful [*kalos*] and bad [*kakos*]). One branch appeared healthy and eugenic, descended from Martin's marriage to a "respectable" woman. The second branch was composed of "Defective degenerates" (alcoholics, criminals, prostitutes, and particularly the mentally "feebleminded") born of Martin's dalliances with a "feebleminded" tavern mistress. Goddard thus "proved" the inheritance of feeblemindedness, and its social cost.

Convinced that "feeblemindedness" and other complex antisocial behaviors behaved like simple Mendelian traits, eugenicists lobbied for compulsory sterilization laws. Between 1907 and the mid-1930s, such laws were adopted by thirty-two American states. The U.S. Supreme Court upheld these laws in 1927, when Justice Oliver Wendell Holmes ruled that, "the principle that sustains compulsory vaccination of schoolchildren is broad enough to cover the cutting of the fallopian tubes. . . . Three generations of imbeciles are enough." "Feebleminded" individuals were prominent

People gather for the first Fitter Family contest, sponsored by the American Eugenics Society, at the Kansas State Free Fair in 1920. At the height of the Eugenics movement's popularity such exhibits were well attended—people participated in testing meant to evaluate their "eugenic fitness." Winners of such contests were white, with Northern and/or Western European heritage.

among the more than 60,000 individuals sterilized in the United States under eugenic sterilization laws between 1927 and 1979.

Nazi Eugenics

Eugenics is commonly associated with the Nazi racial hygiene program that began in 1933 and ended in May 1945, with Germany's defeat near the end of World War II. Although the German eugenics movement existed long before the Nazis came to power, scholars have shown that Nazi eugenicists were inspired by American eugenic studies and sterilization, as well as their antimiscegenation and immigration restriction laws.

Goddard's *Kallikak* study was well respected among German eugenicists who, like American eugenicists, emphasized genetics as the basis of human differences. German racial hygienists also praised the American eugenicist Madison Grant's racist book, *The Passing of the Great Race*, which Adolf Hitler referred to as his "Bible." The 1933 Nazi "Law for the Prevention of Genetically Diseased Progeny," relied on American examples, especially the model law drafted by American eugenicist Harry Hamilton Laughlin. In 1936 the University of Heidelberg awarded Laughlin an honorary degree for his contributions to "racial hygiene." Laughlin's degree was ordered and signed by Hitler.

The Nazis instituted state-supported positive eugenics programs that encouraged "racially fit" women to reproduce, as well as a massive negative eugenics program that included euthanasia. Ultimately, the Nazis sterilized about 400,000 people and euthanized another 70,000 individuals that were judged to be feebleminded or otherwise unfit. The euthanasia program foreshadowed the extermination of six million Jewish victims, along with millions of others, notably Gypsies and homosexuals, in the Holocaust.

Demise of Eugenics

Early on, some scientists objected to the eugenicists' insistence that heredity overwhelmed environmental influences in shaping human life. Others objected to the eugenicists' methodologies, noting that family studies often relied on hearsay evidence and biased observation rather than direct, quantifiable, empirical measurements. Others challenged the eugenicists' reliance on **phenotypic** traits such as body form to diagnose presumed underlying genetic causes. Developments in genetics increasingly undermined this simplistic reasoning from phenotype to genotype.

phenotypic related to the observable characteristics of an organism

Instead, genetic studies increasingly revealed the complex nature of most human phenotypic traits. Human traits rarely result from the action of single gene pairs, and expression depends on complex environmental influences. Moreover, many genes induce pleiotropic effects: that is, a single gene may influence more than one phenotypic characteristic. If multiple genes cause single traits, or if single genes are involved in many effects, then any attempt to "breed out" traits becomes virtually impossible. Moreover, if, as most eugenicists believed, negative traits are recessive factors in single-gene disorders, then most "bad" genes are harbored in apparently normal, **heterozygous** carriers. The Hardy-Weinberg theorem, formulated in 1908, made it clear that eugenic selection directed solely against affected individuals would barely reduce the incidence of a trait in the larger population. To decrease such defects by half would require forty generations (1,000 years) of perfect negative eugenics.

heterozygous characterized by possession of two different forms (alleles) of a particular gene

The Hardy-Weinberg theorem alone, unfortunately, did not dissuade most geneticists from eugenics. Many continued to believe, as geneticist Herbert Spencer Jennings wrote in 1930, that preventing the "propagation of even one congenitally defective individual puts a period to at least one line of operation of this devil. To fail to do at least so much would be a crime." Nevertheless, the most bigoted aspects of eugenics dwindled after 1946, as scientists recoiled from the horrors of Nazi atrocities.

The dream of improving human life through genetic intervention remains with us today. While genetic knowledge and technology have changed since the Holocaust, the cultural and political context surrounding the pursuit of genetic improvement has undergone even greater transformations. The goal of present genetic intervention is not group improvement, but individual therapy. Modern conceptions of individual, patient, and human rights reduce the risk of abuses committed in the name of eugenics. While negative eugenics has been largely eliminated as neither possible nor socially acceptable, positive eugenics are still considered desirable among some people who propose genetic engineering for the development of children with superior traits. The ethical issues surrounding genetic engineering and cloning are still debated in light of the history of the eugenics movement. SEE ALSO Cloning: Ethical Issues; Gene Therapy: Ethical Issues; Hardy-Weinberg Equilibrium; Inheritance Patterns; Reproductive Technology: Ethical Issues.

Gregory Michael Dorr

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Evolution, Molecular

All life on Earth is cellular and uses DNA to store genetic information. However, evidence suggests that, on ancient Earth, much complex chemical activity preceded cellular development, and it was probably not DNA-based at the start. What was the nature of this activity, and how did it lead to life? "Molecular evolution" is a term used to describe the stages that preceded the origin of life on Earth. The term implies that information-containing molecules were subject to the process of natural selection, whereby genetic structures were capable of both replication (the copying of specific **nucleic acid** sequences) and **mutation**. In addition, it is theorized that certain chemical reactions may have taken place on early Earth, before true evolution began, and some of these reactions may have helped to form these informational molecules that enabled **replication** and mutation.

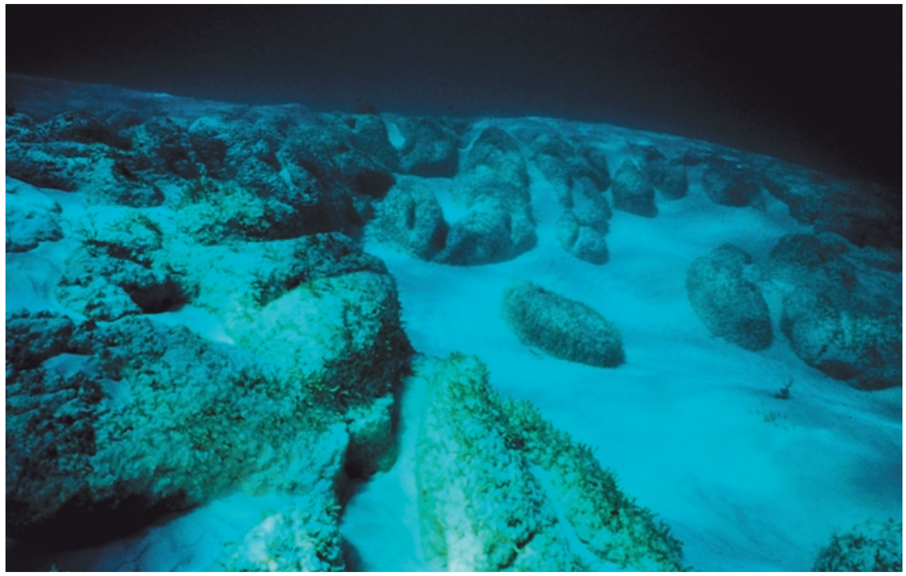
nucleic acid DNA or RNA

mutation change in DNA sequence

replication duplication of DNA

The Antiquity of Life

The oldest known fossils date from about 3.5 billion years ago, and have been found in Western Australia. Called stromatolites, these domelike rocks



Stromatolites, like these in the shallow waters of the Bahamas, have existed for over three million years. These rocks contain fossils of various cyanobacteria such as colonial chroococcalean forms, and filamentous *Palaeolyngbya*.

are formed today by photosynthetic organisms (cyanobacteria) that live in shallow waters. Large communities of cyanobacteria form microbial mats that trap sediment, providing a sturdy foundation on which another layer of bacteria can grow.

The search for older fossils has been frustrated by the scarcity of pristine rocks that have not been altered by high temperatures and pressures. However, evidence from geological deposits in Greenland dated at 3.87 billion years ago may point to the presence of life at an even earlier time. Since our planet itself is approximately 4.6 billion years old, and several hundred million years were needed for the surface to cool to "hospitable" temperatures, the origin of life must have been quite rapid compared to the vast span of terrestrial history.

Building the Building Blocks: RNA Nucleotides

A productive hypothesis, which has stimulated the design of laboratory experiments that might mimic the formation of biochemical compounds on early Earth, has been the notion that ribonucleic acid (RNA) was the primordial genetic material. Many scientists believe that RNA arose *before* DNA, and that DNA later took over the information-storage role from RNA. Among the reasons for this view are the modern routes by which pieces of DNA are made. For example, the characteristic sugar that forms the backbone is generated by an **enzyme** that removes an oxygen (the "deoxy" in deoxyribonucleic acid) from the corresponding RNA sugar, prior to assembly of the chain. This enzyme probably evolved after the appearance of RNA components on early Earth.

enzyme a protein that controls a reaction in a cell

RNA chains are composed of repeating units called **nucleotides**. Each nucleotide consists of a phosphate and a ribose sugar, to which one of the four "bases" (uracil, adenine, cytosine, and guanine) is appended. While phosphate minerals were common on Earth early in its history, each of the other components has been the target of laboratory simulations to determine how they might have been formed.

nucleotides the building blocks of RNA or DNA

The bases appear to have been relatively easy to create. Hydrogen cyanide, a possible ingredient in the early oceans and lakes, reacts in the presence of ultraviolet light to give off adenine. Other cyanide derivatives (along with urea) can produce the other bases. A reasonable natural setting for this chemistry would have been a shallow lake or lagoon, as envisioned by Stanley Miller in his preparation of cytosine.

More challenging to those studying the origins of life has been determining how ribose may have been formed and to explore how it may have reacted with the bases (especially cytosine and uracil) and acquired the phosphate to form RNA. Formaldehyde (the same chemical used to preserve specimens) is a likely **prebiotic** molecule that reacts with itself to give a very complex mixture of products that includes traces of ribose. A more effective route, however, starts with formaldehyde and a derivative known as glycolaldehyde phosphate: these react under alkaline conditions to give mainly a ribose compound with two phosphates attached, and the reaction is promoted by certain minerals.

prebiotic before the origin of life

Heating a mixture of ribose with either adenine or guanine to dryness results in some bond formation between the base and the sugar, but this strategy has failed when cytosine or uracil was used in place of adenine and guanine. More research is still needed to establish plausible routes to these RNA precursors on early Earth, and some skeptics have even proposed that a simpler type of backbone may have preceded that found in nucleic acids today.

Linking Subunits into Chains

Mineral catalysts have also been useful in forming chains of RNA-like molecules from the activated precursors. James Ferris has focused on the ability of montmorillonite (a type of clay) to promote the assembly of the ribose-phosphate backbone, starting with a special, uracil-containing compound known as a phosphorimidazolide. Although phosphorimidazolides do not occur in nature today, they are highly reactive species and closely related to the modern building blocks of RNA. Most scientists do not maintain that these compounds were present in the **primordial soup**, but they are convenient substitutes for the natural precursors of RNA.

primordial soup hypothesized prebiotic environment rich in life's building blocks

When the adenine derivative of this compound binds to the mineral surface, the products include chains of up to ten units long, primarily with the "biologically correct" bonds between adjacent riboses. By repeated additions of the starting material, the process can extend the structure up to fifty units. The uracil derivative reacts in a similar fashion, although the chains are somewhat shorter. These data suggest that binding to mineral surfaces may have been important in controlling the proximity and orientation of molecules that could give rise to the first RNA-like fragments, and set the stage for subsequent replication.

RNA Replication without Enzymes

Once formed, how would the first RNA chains cause copies of themselves to be created? Replication is the process by which DNA or RNA makes copies of specific base sequences. However, the "copy" is not identical to the original, called a template. Rather, it is analogous to a photographic negative, with predicable differences from the original. In the case of RNA, the

hydrogen bonding weak bonding between the H of one molecule or group and a nitrogen or oxygen of another

template a master copy

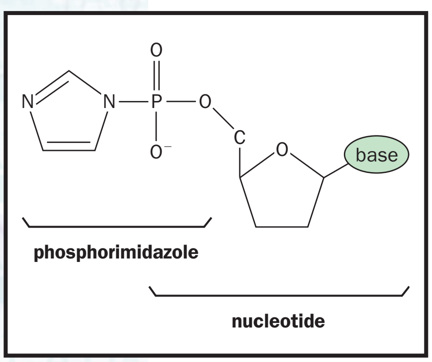
enzymes proteins that control a reaction in a cell

new strand has a different pattern of bases, determined by the specific interactions (**hydrogen bonding**) between the old strand and the new one. Thus, adenine bonds to uracil (causing it to become part of the copy), and guanine "directs" the incorporation of cytosine in the same way. As an example, a **template** sequence abbreviated as AACCAA would be replicated as UUGGUU (the letters stand for each of the four bases). Of course, the faithful transfer of genetic information is a much more complex process, involving an array of complicated protein enzymes and requiring special "activated" precursors for each of the bases.

Leslie Orgel and his associates have carried out extensive studies of replication since 1980. Their goals were, first, to establish whether this process could occur without the help of **enzymes** (which would not have been present in the environment of early Earth); second, to analyze the accuracy of the copies; third, to explore the limits on the types of sequences that could be copied; and fourth, to determine if the copies could then serve as templates for self-perpetuating replication. These aspects have met with different degrees of success, as discussed below.

Orgel's first breakthrough came with the reaction of activated guanine derivatives (again using phosphorimidazolides) on a template consisting of repeating cytosines, which was thus very similar to a small piece of RNA, except that it had only one type of base. In the presence of a zinc catalyst, the guanine derivatives bound to the template and formed chains with more than thirty guanines linked to one another. (Without the template, the only products were those with two or three bases.) Equally striking was that the guanine chain contained mainly the same type of phosphate-ribose backbone as in native RNA, and the template preferentially bound the "correct" base more than 99 percent of the time. Even if the activated derivatives of uracil, adenine, and cytosine were present, they become incorporated into the product with less than 1 percent efficiency. These early experiments demonstrated that templates could accurately form long chains with the appropriate bond between the sugar and phosphate.

The copying of other sequences using this approach has been more difficult, however, partly because of the unreactive structures that the templates often form. For example, RNA chains containing only adenine mixed with other chains of uracil tend to form aggregates that hinder replication. Guanine is even more unusual, in that it organizes into arrays with four chains locked together, which also prevents replication, although this problem may be overcome by employing very dilute reaction conditions (more relevant to early Earth). The greatest success has been achieved with mixed, cytosine-rich templates that contain adenine, guanine, or uracil as isolated bases separated by at least three cytosines.



The phosphorimidazole activates the nucleotide, promoting the linking of multiple units. This synthetic compound may mimic compounds present in prebiotic environments.

A further difficulty is that none of the systems studied by Orgel is capable of replication beyond the first stage: the copies can never serve as templates themselves, because they remain tightly bound to the original template. However, Gunter von Kiedrowski made an important advance in 1994, when he showed that sets of three bases containing guanine and cytosine on a DNA backbone could assemble on a template six units long and then separate. For example, two fragments of GGC could link together on a CCGCCG framework, and then break apart to form a GGCGGC template, available for further replication. The reaction conditions were quite different from what might have existed on early Earth, especially the chemical used to form the bond between the GGC units, but it supports the concept that true replication of this type might be possible.

RNA Can Act as an Enzyme

In cells, replication is controlled by protein **catalysts** called enzymes. However, since proteins are thought not to have been present on early Earth, how could replication and its related reactions been catalyzed? A key discovery, which has affected how molecular biologists and biochemists view RNA, is that it can also act as a catalyst. In the early 1980s Thomas Cech and Sidney Altman independently discovered RNA (that is, non-protein-based) catalysts in a variety of cell types, whereas scientists had previously thought that only proteins have this power. The concept that RNA could both store information and accelerate biochemical processes made it a much more likely candidate for catalyzing reactions in early life.

catalysts substances that speed up a reaction without being consumed (e.g., enzyme)

The catalytic RNAs (known as **ribozymes**) found in nature today mainly promote reactions that involve removing certain sequences from an RNA chain before it can be used in protein formation. However, a variety of exotic tools in the molecular biologists' arsenal have now allowed the creation of new ribozymes by **in vitro** evolution (evolution in a test tube), a method developed by Jack Szostak. He and his associate, David Bartel, showed in 1993 that a novel ribozyme with the ability to link together two RNA chains could be evolved in the laboratory through successive rounds of selection and amplification (creating many copies of the most effective sequence from the previous round). After eight rounds, the best catalyst was faster by a factor of three million, compared to the uncatalyzed reaction.

ribozyme RNA-based catalyst

in vitro in glass"; in lab apparatus, rather than within a living organism

Bartel has applied this strategy toward the development of the first RNA that catalyzes replication: it binds a smaller RNA template, which then creates a copy that is up to 14 units long. When a set of these replicated products was analyzed, the average accuracy for incorporation of the correct base was 96.7 percent (lowest accuracy was achieved for adenine, greatest for guanine). Although the ribozyme itself was 189 units long and thus represents a highly complex structure that would probably not have formed spontaneously in the early ocean, the experiment demonstrates that RNA can promote the critical step of "peeling off" the copy to allow another cycle of product formation. Such evolutionary exercises thus provide a powerful method for exploring the relationship between ribozyme sequence and the catalytic activity.

Goals for Future Research

Theories of molecular evolution have been based on the paradigm that RNA (or a similar structure) served as the first genetic molecule. Many gaps remain in our understanding, from the assembly of the individual units, to the replication of RNA sequences. Further advances in the study of catalysis, especially with minerals and ribozymes, may provide new avenues for future researchers to explore. SEE ALSO Evolution of Genes; Nucleotide; Ribozyme; RNA Processing.

William J. Hagan

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Evolution of Genes

genome the total genetic material in a cell or organism

Individual genes and whole **genomes** change over time. Indeed, evolution of genes ultimately accounts for the evolution of organisms that is seen in the fossil record: Humans evolved from earlier apes, and those creatures from their ancestors, by gene changes in the earlier creatures. Just as the fossil record can be examined to understand the patterns of organismic evolution, so too can genes be compared to understand genomic evolution.

Natural Selection

heritable genetic

Most changes that occur in genes are subject to natural selection, the process first outlined by Charles Darwin in 1859. In natural selection, a **heritable** change arises by chance. If the organism with that change is better able to survive and reproduce, it will leave more descendants in future generations. These descendants will also carry the new genetic change, and as they reproduce, the change will become more widespread in the population. On the other hand, if the change decreases an organism's survival rate, it will be lost from the population. It is also possible to have a neutral change, with no immediate effect on survival. Such "hidden" genetic variation within a population provides grist for evolution when it offers a selective advantage under new environmental conditions.

It is important to remember that a genome (all the DNA of an organism) is more than just its genes. The genome includes vast amounts of DNA outside of genes, and this too is subject to change over time. In fact, nongene portions usually change at a faster rate than genes, because many of these changes have little or no effect on the organism's survival.

Evolution of genes and genomes includes sequence changes to existing genes, gene duplication, recombination of gene segments, and the varied actions of transposable elements as they move through the genome.

Point Mutations in Existing Genes

amino acids building blocks of protein

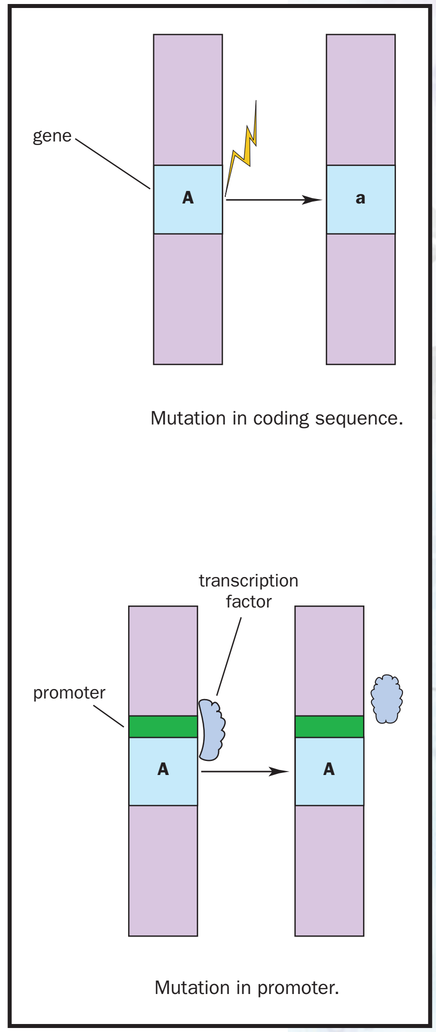
Genes are long strings of four nucleotides (abbreviated A, T, C, and G) whose order dictates the order of **amino acids** in proteins (or nucleotides in RNA). A point mutation is a change in a single nucleotide position at a particular point in a gene. Point mutations can be changes that convert one type of nucleotide to another (a C to a T, for instance), or cause the deletion or addition of a single nucleotide. While the rate of mutation is slow, over long periods (millions of years) most possible sequence changes will have occurred several times in a population, and so natural selection is likely to have acted on most genes in almost every modern population.

Some point mutations can change the amino acid sequence of the resulting protein, altering its properties. For instance, a digestive enzyme's attraction for its substrate (the food molecule it breaks down) may be altered to allow the organism to digest new foods, or prevent it from digesting old ones. The sickle form of hemoglobin arose because of a point mutation that changed a single amino acid in hemoglobin. The result was a molecule that bound oxygen less tightly under certain conditions, conferring resistance to malaria, but also causing sickle cell disease, a type of hemoglobinopathy.

Other point mutations may leave the protein unchanged, but alter the conditions under which it is expressed. Genes are expressed (that is, are "read" to cause protein production) when **transcription factors** bind to a region of the gene known as the promoter. Promoters interact with other DNA regions, called enhancers, which are often a long distance from the gene along the chromosome, but are nonetheless close to it because of folding of the DNA. Mutations in either the promoter or enhancer region can have profound effects on the sensitivity of expression to hormones, temperature changes, and other regulatory influences. For instance, a human gene coding for one form of an enzyme called pancreatic elastase appears to have been silenced by an evolutionarily recent enhancer mutation.

Not all DNA sequence changes will lead to changes in the encoded protein or in the way that it is expressed. For many amino acids, there are several DNA "synonyms" that all code for the same amino acid. GGG, GGC, GGA, and GGT all code for the amino acid proline, for example. This is known as the degeneracy of the genetic code. Also, mutations that code for chemically similar amino acids may have no significant effect on protein function. For instance, leucine (AAT) and isoleucine (TAT) are both small, nonpolar amino acids often found on the interior of proteins. Mutations that exchange one for the other may have little effect on protein structure or function.

Genes in **eukaryotes** also contain noncoding regions, called introns. Mutations in these regions often have no effect, since their sequences do not code for part of the finished protein. Changes to the ends of introns and certain internal sequences may have an effect, however, since these control the removal of the intron sequence from the RNA copy after transcription. A mutation here can prevent intron removal, altering the finished protein. One form of the hemoglobin disease beta-thalassemia is due to an intron mutation.



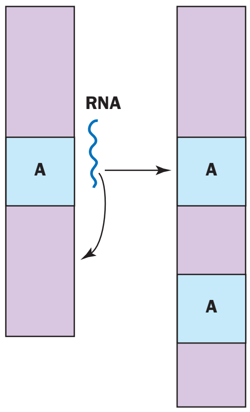
Mutations in the coding region may change the protein that results. Mutations in the promoter can affect how transcription factors bind, altering the level of gene expression.

transcription factors proteins that increase the rate of gene transcription

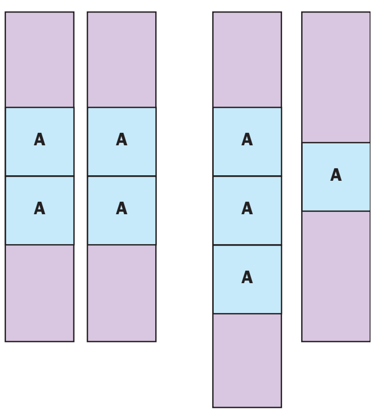
eukaryotes organisms with cells possessing a nucleus

Alleles

Humans and most other multicellular creatures are diploid, meaning they carry two sets of virtually identical chromosomes (one inherited from the mother, one from the father). The two members of a chromosome pair (called homologous chromosomes) carry identical sets of genes, so that each



Duplication by retroposition.



Amplification during crossing over.

Two common mechanisms of gene duplication.

alleles particular forms of genes

organism has two copies of each gene. A point mutation in one of these creates a new form of the gene. Different forms of the same gene are called **alleles**. Creation of alleles is one of the most common forms of gene evolution. The existence of diploidy allows a greater tolerance of new alleles, since a mutation to one allele still leaves the other one functioning, and in many cases this may be sufficient for survival. The existence of alleles allows greater genetic diversity in a population, which may increase that population's ability to adapt to changing environmental conditions.

Gene Duplication

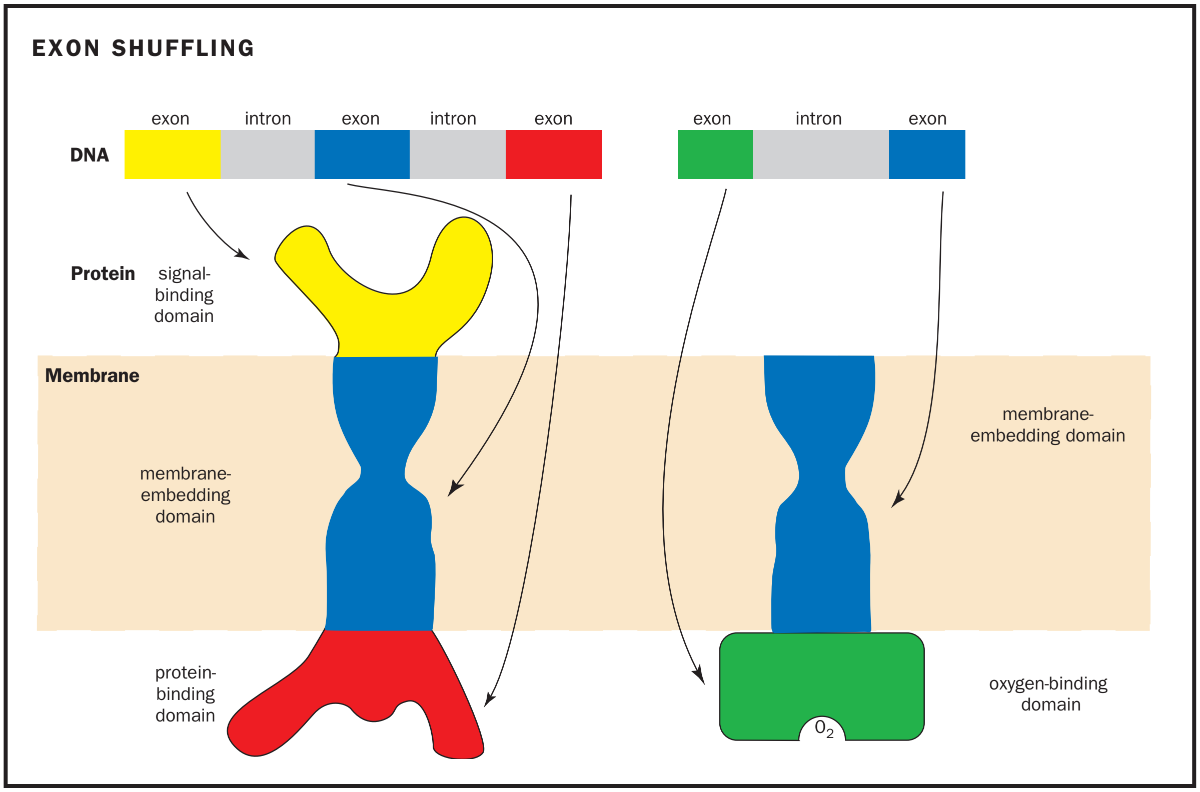
Occasionally a gene on a single chromosome will be duplicated to create a pair of identical genes. Duplication may occur for any one of several reasons. One type of duplication occurs when the RNA transcript of a gene is "reverse transcribed" back into a DNA sequence and reinserted elsewhere in the genome (a process called retroposition), leading to a new, possibly functional copy of the gene in a new location and subject to different regulatory systems. This appears to have occurred with the human gene for phosphoglycerate kinase 2, which is involved in energy use in the cell.

More likely, a retroposed gene will be functionless, since it will not have the promoters and enhancers it needs for expression. Typically, one copy of a duplicated gene is either nonfunctional or accumulates mutations that render it so. After a long period of evolutionary time, the duplicate gene may acquire so many mutations that it may be difficult to see its relationship to its parent gene. Nonfunctional copies of previously functional genes are called pseudogenes.

Analysis of genomes shows that many gene copies are found lying next to each other, linked head to tail in an arrangement called a "tandem repeat." This may occur because of errors of the normal recombination machinery that is responsible for DNA repair and crossing over during meiosis. Tandem repeats are susceptible to amplification, which is the further increase in the number of copies. This can occur during crossing over. Normal crossing over pairs up identical segments on homologous chromosomes, and then exchanges them. If the chromosomes each have a tandem repeat, the crossover machinery may line up incorrectly, leaving one homologue with three gene copies and one with only one. Repeating this process over ensuing generations can lead to dozens of extra gene copies.

Duplication of much larger portions of a genome is also possible, including whole chromosomes (called chromosomal aberrations) and even the entire genome (called polyploidy). In each case, the number of copies of a gene increases. Such copies are usually removed by natural selection, but it is sometimes advantageous to have several gene copies, particularly for those genes that code for ribosomal RNAs. These are present in dozens or even hundreds of copies, allowing rapid production of new ribosomes during cell growth.

While gene duplication is a rare event in the short term, it is frequent enough in the long term to have been a central feature in the evolution of the genome of eukaryotic organisms. As with alleles, gene duplication frees up a gene copy to accumulate mutations with less selective penalty. Over time, such a gene may change its function slightly, or even acquire a new



function, that increases the capabilities of the organism. The modification of duplicated genes therefore provides the diversity that is acted on by natural selection. An increase in the rate of gene duplication is thought to be one factor contributing to the diversification of multicellular organisms during the so-called Cambrian explosion 600 million years ago, which gave rise to most of the basic animal forms that exist today.

The "exon shuffling" model of protein evolution suggests that exons encode functional domains of protein, and can be shuffled to create proteins with novel functions.

Gene Families

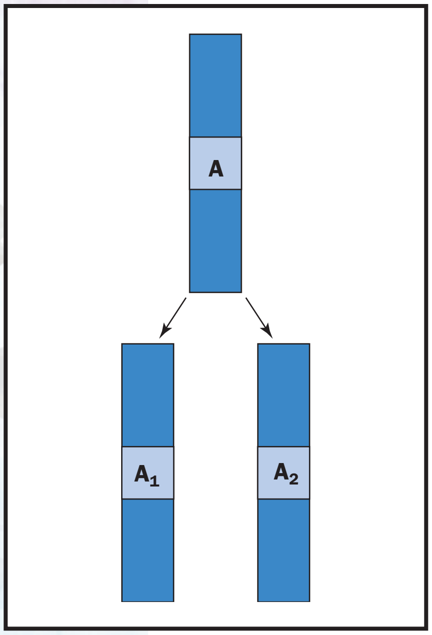
The set of genes that evolve from a single ancestral gene comprise a gene family. In humans, gene families are found in many important groups of proteins, and more are being discovered as the human genome is explored. These include the globins, which carry oxygen in the blood (hemoglobin) and store it in muscle (myoglobin); the immunoglobulins (specifically, the heavy chain of the immunoglobulin), which form antibodies of the immune system; the actins, which move the cell cytoskeleton and muscle; the collagens, which form cartilage and other structural materials; and the homeotic genes, master controllers of embryonic development. There are many other examples as well.

A new member of a particular gene family is discovered by comparing its sequence to known members. This is usually done by computerized database search, and is one of the challenges of **bioinformatics**, a new specialty devoted to collecting and analyzing large amounts of biological information.

bioinformatics use of information technology to analyze biological data

Pseudogenes

Over time, some gene copies mutate to lose their function entirely. Such so-called pseudogenes may arise through accumulation of mutations that



A gene family is descended from a common ancestral gene.

translation synthesis of protein using mRNA code

prevent translation of the gene, such as an insertion or deletion that stops **translation** at the beginning of the gene sequence. Pseudogenes also arise from mutation in a gene's promoter region. The promoter is the site at the beginning of the gene that attracts the enzyme called RNA polymerase. Without a functional promoter, the gene cannot be transcribed effectively, and so cannot lead to protein production.

Retroposition is a very common source of pseudogenes. Pseudogenes have been discovered because their sequences are similar to functional genes. In humans, pseudogenes are known to exist for topoisomerase (a gene that cuts DNA to prevent twisting), ferritin (an iron storage protein), two different forms of actin, and many other genes.

The Role of Transposable Genetic Elements

Transposable genetic elements are DNA segments that move around in the genome. Many biologists consider them to be a form of "selfish DNA," a kind of genetic parasite that serves no useful function for the host, but remains in the genome because it is efficient at getting itself copied. They can be present in large numbers of copies. In humans, more than a million copies of a single element, called Alu, account for about ten percent of the entire genome.

Insertion of a transposable element can disrupt a gene, creating a pseudogene. When a transposable element moves, it occasionally also takes a gene with it, placing it in a new position under the control of different regulatory elements. Alternatively, it may move an enhancer, thus affecting both the gene whose enhancer was removed and the gene (if any) it is now placed closer to. Some transposable elements themselves contain enhancers, further increasing the chances of altering gene expression when they are inserted in a new location.

Exon Shuffling

The coding portions of eukaryotic genes, termed "exons," are interrupted by noncoding regions, termed "introns." The evolutionary role of introns has been controversial since their discovery in 1977. Some scientists propose they are just another form of "junk DNA," and may be the relics of transposable elements or other forms of selfish DNA. Others suggest they may have played a central role in protein evolution.

The argument about the evolutionary importance of introns turns on exactly how they divide up the genes in which they are found. Proteins, which are encoded by genes, are not random strings of amino acids, but rather highly organized three-dimensional shapes, with different functions served by discrete parts, known as domains. A protein may contain half a dozen domains; one may bind a signaling molecule from outside the cell, another embeds the protein in a membrane, another binds an internal protein, and so on. It is often the case that each domain in a protein is folded up from a discrete segment of the amino acid chain.

Just as the domain's amino acids occur in sequence in the protein, the nucleotides that code for them occur in sequence in the gene. Those who propose that introns play a vital role in protein evolution suggest that exons correspond to the protein's domains, and that introns serve to divide the gene into these useful little bits of code. In this view, exons serve as "modules," or useful gene segments, that can be shuffled (via gene duplication and transposable elements, for instance) to create genes for new proteins with novel functions. For instance, a module for a membrane-embedding domain could be linked to a module for an oxygen-binding domain, allowing oxygen to be stored on a membrane, or a hormone-binding domain might be joined to a promoter-binding domain, allowing a hormone to control gene transcription.

One form of the disorder hemophilia is due to the insertion of a transposable element into a blood clotting gene.

The validity of this model of protein evolution depends on whether a gene's exons do indeed correspond to its protein's domains, and whether introns do actually separate domain-coding regions. So far the evidence is mixed, with some genes clearly divided this way, but many others showing complex or conflicting structures.

Because of this, scientists do not yet agree on the importance of exon shuffling in protein evolution. While it likely has occurred, it is unknown how widespread it may be. Also at issue is whether introns themselves arose early or late in life's evolution. If early, it may have been central to the development of all forms of life. The absence of introns in bacteria would then presumably be due to a streamlining of their genome by natural selection. If introns arose late, they were probably confined to eukaryotes and were therefore only important in their evolution.

While there is much that remains controversial, there is little disagreement about the importance of a related use of exons that occurs continually in many tissues. This is called alternative splicing. In this case, particular exons may be omitted, or they may be reassembled differently from tissue to tissue, creating tissue-specific variants, called isoforms, of the same protein. SEE ALSO Alternative Splicing; Bioinformatics; Chromosomal Aberrations; Development, Genetic Control of; Gene; Gene Families; Genetic Code; Hemoglobinopathies; Immune System Genetics; Mutation; Polyploidy; Pseudogenes; RNA Processing; Transposable Genetic Elements.

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Extranuclear Inheritance *See Inheritance, Extranuclear*

Eye Color

When someone asks, "What color are your eyes?" he should really be asking "What color are your irises?" because it is the iris that contains the pigment that determines the color of your eyes. Despite the fascination eye color holds for us, the genes responsible for it in humans are not well-known and are more complex than most people think.